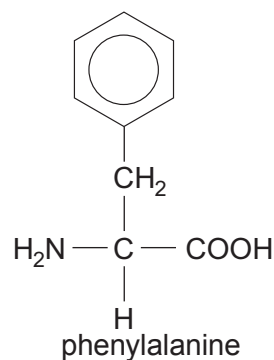
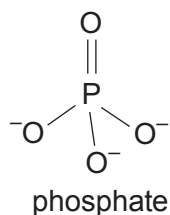
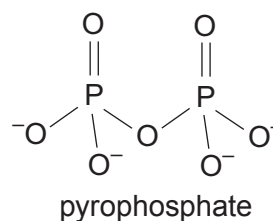
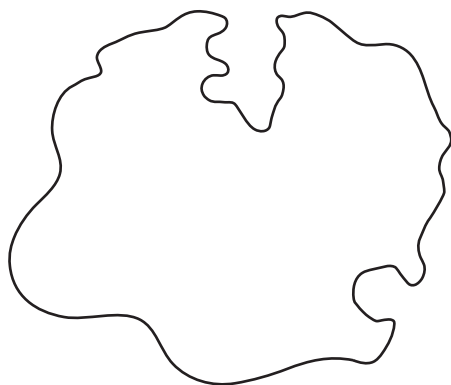


6. Pyrophosphatase is an enzyme found inside the nucleus of cells and is involved in DNA replication. The enzyme catalyses the conversion of a molecule of pyrophosphate to two phosphate ions. The diagrams below show the enzyme pyrophosphatase and its substrate pyrophosphate. Molecules of phenylalanine (an amino acid) and phosphate are also shown; both of these molecules are known to inhibit pyrophosphatase. (Drawings are not to the same scale).



Describe and explain why pyrophosphatase can only hydrolyse pyrophosphate and the mechanism by which phenylalanine and phosphate inhibit pyrophosphatase. [9 QER]



Examiner
only

Handwriting practice area with horizontal dotted lines.



Examiner
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9

END OF PAPER

